

1.1 Systems architecture

Where this unit sits, what it covers, and how far

Where this unit sits

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component	Component 01: Computer systems — 80 marks, 1 hour 30 minutes
What OCR says the component covers	the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software
This unit	topic 1.1 Systems architecture
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

Before it: nothing — this is the first unit of the course

After it: 1.2 Memory and storage, 1.3 Computer networks, connections and protocols, 1.4 Network security, 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Fetch-execute factory sa-101	What is the architecture of a CPU?	12	22
2	Von Neumann HQ sa-102	What other architectures exist and where are they used?	15	34
3	Speed lab sa-103	What factors affect CPU performance?	13	17
4	Smart home sa-104	What are embedded systems, and what are their characteristics?	12	24
5	End-of-topic test sa-105-test	assesses the whole unit	0	0

Prior knowledge, vocabulary and skills

Assumed before the unit: nothing beyond the earlier units of this course

Vocabulary introduced: CPU, Fetch-decode-execute cycle, ALU, CU (control unit), Cache, Register, Von Neumann architecture, MAR

What a pupil does, not only reads: match, mcq, multi, order, sort

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in `build/coverage.json`. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand what the CPU of a computer does <code>sa-l01-o1</code>	sa-l01	yes	yes	yes	Full
know what CPU registers are <code>sa-l01-o2</code>	sa-l01	yes	yes	yes	Full
know the stages of the fetch-execute cycle <code>sa-l01-o3</code>	sa-l01	yes	yes	yes	Full
know about other components of the CPU <code>sa-l01-o4</code>	sa-l01	yes	yes	yes	Full
describe the von Neumann architecture <code>sa-l02-o1</code>	sa-l02	yes	yes	yes	Full
know the components of the von Neumann architecture <code>sa-l02-o2</code>	sa-l02	yes	yes	yes	Full
understand what a keyword is <code>sa-l02-o3</code>	sa-l02	yes	yes	yes	Full
understand three ways in which the speed of a CPU, and therefore a computer, can be increased <code>sa-l03-o1</code>	sa-l03	yes	yes	yes	Full
know the units clock speed is measured in, and what overclocking does to heat <code>sa-l03-o2</code>	sa-l03	yes	yes	yes	Full
understand what is meant by the term embedded system <code>sa-l04-o1</code>	sa-l04	yes	yes	yes	Full
know several examples of embedded systems <code>sa-l04-o2</code>	sa-l04	yes	yes	yes	Full
understand how an embedded system takes input, processes it and produces output <code>sa-l04-o3</code>	sa-l04	yes	yes	yes	Full

12 full · **0 partial** · **0 not covered**, of 12 outcomes.

Gaps, assumptions and overlap

None found by the trace.

Assessed, but matching no outcome this unit states

Each of these is a real exam question on a worksheet in this unit. Either the outcome it tests is worded differently, or it is genuinely assessing something the unit does not claim to teach.

- sa-101-x1 Describe what the CPU does while a program is running.
- sa-101-x3 A computer's cache is full. Explain the effect this has on the CPU.
- sa-103-x2 A laptop is advertised with a quad-core 3.2 GHz processor. Explain what this tells you about the CPU.

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